

EDUCATION

University of Maryland

M.Eng. in Robotics

AUG 2024 - MAY 2026

COLLEGE PARK, MD

TECHNICAL SKILLS

Programming & Development: Python, C++, ROS2, Git, Linux, Embedded C

Machine Learning & AI: PyTorch, JAX, TensorFlow, Reinforcement Learning (PPO, TD3), Imitation Learning, NumPy, SciPy, scikit-learn, OpenAI Gym

Motion Planning & Control: A*, Dijkstra, RRT, RRT*, PRM, Control Systems (PID, LQR/LQG), MoveIt

Robotics Hardware: Universal Robots (UR3e, UR5e, UR10e), Franka Emika Panda, Turtlebot 4

Simulation : Gazebo, RViz, PyBullet, MuJoCo, CARLA, SAPIEN, NVIDIA Isaac Sim

EXPERIENCE

Artly AI

Robot Learning Intern (via UMD)

JAN 2026 - PRESENT

SEATTLE, WA (REMOTE)

- ▶ Building dexterous manipulation pipelines in MuJoCo and on hardware using the RealHand L6, focusing on grasp synthesis, policy learning, and multimodal perception.
- ▶ Developing data collection and evaluation frameworks for large-scale manipulation datasets to improve grasp stability and generalization.

RAAS Lab - Maryland Robotics Center

Research Assistant

JAN 2025 - PRESENT

COLLEGE PARK, MD

- ▶ Currently developing a transfer learning framework that infers action semantics at inference time, enabling single RL agents to operate across heterogeneous state-action spaces, tested on OpenAI Gym Retro and multi-DOF robotic embodiments.
- ▶ Deployed and fine-tuned Pi-Zero policies on a UR3e for grocery bagging and bussing table tasks, achieving sub-6 ms inference latency, 88% task success, and less than 0.5% damage.
- ▶ Calibrated and integrated a multi-camera perception system with three Intel RealSense depth cameras and Anyskin tactile sensors to enable multimodal feedback for manipulation tasks.

IIT Madras - Guidance, Navigation, & Control Lab

Research Intern

FEB 2024 - JUL 2024

CHENNAI, INDIA

- ▶ Implemented the Gradient Direction Turn Switching (GDTs) algorithm in ROS 2/Gazebo for UAV search-and-rescue source localization, improving trajectory efficiency in dynamic environments.
- ▶ Developed 3D Velocity Obstacles (VO) for multi-drone collision avoidance, enabling safe and reliable path planning in cluttered aerial scenarios.

PROJECTS

PPO and DDPG Based Parallel Parking Agent in CARLA [link](#)

- ▶ Developed and benchmarked deep reinforcement learning agents (PPO/DDPG) on time-series sensor data for autonomous parking; achieved 93% + success rates in dynamic scenarios using 500+ trial dataset.
- ▶ Designed real-time sensor fusion and neural-policy pipelines leveraging LiDAR, radar, and visual streams; applied curriculum learning and reward shaping to reduce maneuver time by 20% while maintaining 95% safety compliance.

Deep Multi-Agent Highway On-Ramp Merging in Mixed Traffic [link](#)

- ▶ Reimplemented and extended the Deep Multi-Agent Reinforcement Learning for Highway On-Ramp Merging framework (Chen et al., arXiv:2105.05701) in OpenAI Gym using PPO; integrated action masking and prioritized safety supervision to boost collision reduction from 20% to 30% and merge speed gains from 10% to 15%.
- ▶ Employed centralized training with decentralized execution and curriculum learning across increasing traffic densities, sustaining greater than 90% merge success under high-volume scenarios.

Autonomous Barista Robot [link](#)

- ▶ Led team of 4 in developing a ROS2 package for a 6-DOF UR10e manipulator, simulating coffee-making processes in a custom Gazebo environment with RViz visualization.
- ▶ Designed position, velocity, and effort controllers that achieved precision in end-effector path following; reduced trajectory deviation to less than 2mm during operation.

Lerobot Hackathon 2025 [link](#)

- ▶ Assembled dual-arm Koch manipulators and developed Python control wrapper for synchronized joint-state feedback, collecting demonstration data via active-passive mimicking for pick-and-place tasks.
- ▶ Trained Behavior Cloning and Action Chunking Transformer (ACT) policies on the Koch arm, achieving a 92% task success rate.